

# Mattia Sensi

Caritro postdoctoral fellow at Università degli Studi di Trento

Università degli Studi di Trento  
Via Sommarive 14, 38123 Povo (Trento) - Italy

Email: [mattia.sensi@unitn.it](mailto:mattia.sensi@unitn.it)  
Website: [mattiasensi.github.io](https://mattiasensi.github.io)

## RESEARCH INTERESTS

Mathematical modelling, mathematical biology, mathematical epidemiology, dynamical systems, billiards, multiple time scales dynamics, Geometric Singular Perturbation Theory (GSPT), qualitative theory of ordinary differential equations, partial differential equations, integro-differential equations, delay differential equations.

## EMPLOYMENT

**Postdoctoral researcher in Mathematics**, Università degli Studi di Trento. March 2025 – present  
**Caritro postdoctoral fellowship**, principal investigator of the project “*Modelli matematici di malattie infettive più ospiti e popolazioni eterogenee: applicazioni all’influenza aviaria*” (Mathematical models of infectious disease spreading in multi-host and heterogeneous populations: applications avian flu); in the group of [Prof. Andrea Pugliese](#).

**Postdoctoral researcher in Mathematics**, Politecnico di Torino. March 2023 – February 2025  
Working in the group of [Prof. Andrea Tosin](#), as part of PRIN 2020 project “*Integrated Mathematical Approaches to Socio-Epidemiological Dynamics*” (No. 2020JLWP23, CUP: E15F21005420006).

**Postdoctoral researcher in Mathematics**, Inria at Université Côte d’Azur. December 2021 – February 2023  
Working in the group [MathNeuro](#), led by [Prof. Mathieu Desroches](#).

**Postdoctoral researcher in Mathematics**, TU Delft. March – November 2021  
Working in the group [NAS](#), led by [Prof. Piet Van Mieghem](#).

## NATIONAL SCIENTIFIC QUALIFICATIONS

- Qualification for Maitre de Conferences in section 26 - Mathématiques appliquées et applications des mathématiques (Applied mathematics and applications of mathematics). From 11/03/2026
- ASN 2023/2025, Abilitazione Scientifica Nazionale alle funzioni di professore universitario di Seconda Fascia nel Settore Concorsuale 01/A3: Analisi Matematica, Probabilità e Statistica Matematica (Scientific Area 01/A3: Mathematical Analysis, Probability and Mathematical Statistics; qualification to become Associate Professor). From 28/02/2025 to 28/02/2037
- ASN 2023/2025, Abilitazione Scientifica Nazionale alle funzioni di professore universitario di Seconda Fascia nel Settore Concorsuale 01/A4: Fisica Matematica (Scientific Area 01/A4: Mathematical Physics; qualification to become Associate Professor). From 07/03/2025 to 07/03/2037

## EDUCATION

**Ph.D. in Mathematics**, *cum laude*, Università degli Studi di Trento. November 2017 – January 2021  
Thesis: “A Geometric Singular Perturbation approach to epidemic compartmental models”  
Supervisor: [Prof. Andrea Pugliese](#)

**M.Sc. in Mathematics**, Universiteit van Amsterdam. September 2015 – June 2017  
Thesis: “Homoclinic vegetation stripes in a Klausmeier-Gray-Scott model”  
Supervisor: [Prof. Dr. Arjen Doelman](#)

**B.Sc. in Mathematics**, Università degli Studi di Padova. September 2011 – September 2014  
Thesis: “Portfolio optimization for quadratic utility function with partial information”  
Supervisor: [Prof. Wolfgang J. Runggaldier](#)

## PUBLICATIONS

- A. Andò, N. Cangiotti and M. S.** *Exploring Exponential Runge-Kutta Methods: A Survey*. Communications in Mathematical Research, 2026, Vol. 42 No. 2, pp. 191-240
- C. Oelen, B. Rink and M. S.** *Non-Birkhoff periodic orbits in symmetric billiards*. Annales Henri Poincare, 2026 (online first). GitHub repository [BilliardOrbitFinder](#)
- L. Eigentler and M. S.** *Wavelength selection for periodic travelling waves: an unsolved problem*. Bulletin of Mathematical Biology, 2026, 88(2), 22
- J. Borsotti and M. S.** *A geometric analysis of the Bazykin-Berezovskaya predator-prey model with Allee effect in an economic framework*. Nonlinear Analysis: Real World Applications, 2026, 89, 104534
- E. Bernardi, T. Lorenzi, M. S. and A. Tosin.** *Heterogeneously structured compartmental models of epidemiological systems: from individual-level processes to population-scale dynamics*. Studies in Applied Mathematics, 2025, 155(2), e70091
- A. Chizhov, L. Pujo-Menjouet, T. Schwalger and M. S.** *A refractory density approach to a multi-scale SEIRS epidemic model*. Infectious Disease Modelling, 2025, 10(3), pp. 787–801
- M. A. Achterberg, M. S. and S. Sottile.** *A minimal model for multigroup adaptive SIS epidemics*. Chaos, 2025, 35(3), 033127
- L. Eigentler and M. S.** *Delayed loss of stability of periodic travelling waves: insights from the analysis of essential spectra*. Journal of Theoretical Biology, 2024, 595, 111945
- I. M. Bulai, M. S. and S. Sottile.** *A geometric analysis of the SIRS compartmental model with fast information and misinformation spreading*. Chaos, Solitons and Fractals, 2024, 185, 115104
- P. Kaklamanos, A. Pugliese, M. S. and S. Sottile.** *A geometric analysis of the SIRS model with secondary infections*. SIAM Journal on Applied Mathematics, 2024, 84(2), pp. 661–686
- R. Persoons, M. S., B. Prasse and P. Van Mieghem.** *Transition from time-variant to static networks: Timescale separation in N-intertwined mean-field approximation of susceptible-infectious-susceptible epidemics*. Physical Review E, 2024, 109(3), 034308

18. M. Adimy, A. Chekroun, L. Pujo-Menjouet and M. S.. *A multigroup approach to delayed prion production*. Discrete and Continuous Dynamical Systems Series B, 2024, 29(7), pp. 2972–2998
17. M. S., M. Desroches and S. Rodrigues. *Slow-fast dynamics in a neurotransmitter release model: delayed response to a time-dependent input signal*. Physica D: Nonlinear Phenomena, 2023, 455, 133887
16. R. Della Marca, A. d’Onofrio, M. S. and S. Sottile. *A geometric analysis of the impact of large but finite switching rates on vaccination evolutionary games*. Nonlinear Analysis: Real World Applications, 2024, 75, 103986
15. N. Cangiotti, M. Capolli, M. S. and S. Sottile. *A survey on Lyapunov functions for epidemic compartmental models*. Bollettino dell’Unione Matematica Italiana, 2024, 17(2), pp. 241–257
14. P. Kaklamanos, C. Kuehn, N. Popovic and M. S.. *Entry-exit functions in fast-slow systems with intersecting eigenvalues*. Journal of Dynamics and Differential Equations, 2025, 37(1), pp. 559–576, 103220
13. N. Cangiotti, M. Capolli and M. S.. *A generalization of unaimed fire Lanchester’s model in multi-battle warfare*. Operational Research, 2023, 23(2), 38
12. M. A. Achterberg and M. S.. *A minimal model for adaptive SIS epidemics*. Nonlinear Dynamics, 2023, 111(13), pp. 12657–12670
11. S. Ottaviano, M. S. and S. Sottile. *Global stability of multi-group SAIRS epidemic models*. Mathematical Methods in the Applied Sciences, 2023, 46(13), pp. 14045–14071
10. N. Cangiotti and M. S.. *Exact solutions for a Solow-Swan model with non-constant returns to scale*. IJPAM, 2023, 54(4), pp. 1278–1285
9. S. Ottaviano, M. S. and S. Sottile. *Global stability of SAIRS epidemic models*. Nonlinear Analysis: Real World Applications, 2022, 65, 103501
8. S. Sottile, O. Kahramanogullari and M. S.. *How network properties and epidemic parameters influence stochastic SIR dynamics on scale-free random networks*. Journal of Simulation, 2024, 18(2), pp. 206–219
7. B. Chang, L. Yang, M. S., M. A. Achterberg, F. Wang, M. Rinaldi and P. Van Mieghem. *Markov Modulated Process to model human mobility*. Complex Networks & Their Applications X. Studies in Computational Intelligence, 2022, 1015, pp. 607–618
6. N. Cangiotti and M. S.. *Benford’s Law: a Number-Theoretical Perspective*. PJM, 2022, 11(3), pp. 379–385
5. N. Cangiotti and M. S.. *A geometric characterization of VES and Kadiyala-type production functions*. Filomat, 2021, 35(5), pp. 1661–1670
4. N. Cangiotti and M. S.. *Notes on a conformal characterization of 2-dimensional Lorentzian manifolds with constant Ricci scalar curvature*. UPB Scientific Bulletin Series A Applied Mathematics and Physics, 2021, 83(2), pp. 129–136
3. T. Lorenzi, A. Pugliese, M. S. and A. Zardini. *Evolutionary dynamics in an SI epidemic model with phenotype-structured susceptible compartment*. Journal of Mathematical Biology, 2021, 83(6-7), 72
2. H. Jardón-Kojakhmetov, C. Kuehn, A. Pugliese and M. S.. *A geometric analysis of the SIRS epidemiological model on a homogeneous network*. Journal of Mathematical Biology, 2021, 83(4), 37
1. H. Jardón-Kojakhmetov, C. Kuehn, A. Pugliese and M. S.. *A geometric analysis of the SIR, SIRS and SIRWS epidemiological models*. Nonlinear Analysis: Real World Applications, 2021, 58, 103220

## PREPRINTS

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4. R. Della Marca, G. Grifò, A. Iuorio and M. S.. *Traveling fronts in a spatial epidemic model with slow loss of immunity*. [Preprint on arXiv](#)
3. J. Borsotti, H. Jardón-Kojakhmetov and M. S.. *Slow-fast dynamics in a planar parasite–host model with an extinction singularity*. [Preprint on arXiv](#)
2. S. De Reggi, A. Pugliese, M. S. and C. Soresina. *A model for mosquito-borne epidemic outbreaks with information-dependent protective behaviour*. [Preprint on arXiv](#)
1. M. Aguiar, B. Kooi, A. Pugliese, M. S. and N. Stollenwerk. *Time scale separation in the vector borne disease model SIRUV via center manifold analysis*. [Preprint on medRxiv](#)

## GRANTS & AWARDS

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- Conference support from [ESMTB](#) (1000€) and from [GNFM](#) (1500€) for the upcoming workshop “[Across generations in mathematical biology](#)” (Trento, October 2026; planned). 2500€
- [Progetti Giovani GNFM 2025](#) with the project “*Analisi geometrica della diffusione spaziale di epidemie su più scale temporali*” (Geometric analysis of the spatial diffusion of epidemics on multiple time scales; CUP E5324001950001), coordinated by Rossella Della Marca, from August 2025 to July 2026. Participant, 2000€
- [Caritro postdoctoral fellowship 2024](#) with the project “*Modelli matematici di malattie infettive più ospiti e popolazioni eterogenee: applicazioni all’influenza aviaria*” (Mathematical models of infectious disease spreading in multi-host and heterogeneous populations: applications to avian flu), from March 2025 to February 2027. Principal investigator, 85000€
- Travel funds from [ESMTB](#) to participate to [ECMTB 2024](#). 200€

## TEACHING EXPERIENCE

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### At Università degli Studi di Trento:

- Teacher for the course [Analisi matematica 1](#), for first year students of Bachelor's Degree in Industrial Engineering, September 2025 – February 2026; September 2026 – February 2027 (planned)
- Teacher for Ph.D. course “Advances in Mathematical Applications to Biology and Medicine: Stability analysis of dynamical systems in mathematical biology”, for first year Ph.D. students in Mathematics, June 2024
- Teaching assistant for [Prof. Alberto Valli](#)'s course *Analisi 1*, for first year students of Bachelor's Degree in Civil, Environmental and Mechanical Engineering, September 2018 – February 2019; September 2020 – February 2021; September – December 2022
- Tutor for [Prof. Andrea Pugliese](#)'s course *Probabilità e Statistica 2*, for second year students of Bachelor's Degree in Biotechnologies, February – May 2018

### At Politecnico di Torino:

- Teaching assistant for [Prof. Luisa Mazzi](#)'s course *Analisi 1*, for first year students of Bachelor's Degree in Aerospace Engineering, October 2023 – February 2024; September 2024 – February 2025

### At Inria – Université Côte d'Azur:

- Teacher of Mathematics for *Linear Algebra Bootcamp*, for first year students of Master's Degree in Computational Neuroscience, September – October 2022

### At Università Popolare Trentina (CFP-UPT):

- Teacher of Mathematics, October 2019 – June 2020

### At Universiteit van Amsterdam:

- Teaching assistant for [Prof. Dr. Rob Stevenson](#)'s course *Numerieke Analyse*, for third year students of Bachelor's Degree in Mathematics, February – June 2017
- Teaching assistant for [Dr. Han Peters](#)' course *Wiskunde 3*, for third year students of Bachelor's Degree in Physics, November – December 2015

### Other:

- Private tutor for [Camplus](#), Torino, May – June 2023
- Private tutor for [WisMon / Bèta onderwijsinstituut](#), Amsterdam and Utrecht, April 2016 – June 2017
- Freelance private teacher of Mathematics and Physics, for high-school and university students, 2008 – present

## MENTORING

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### Master thesis:

- Brian Chang, February – June 2021. [Modeling the Spread of Epidemics](#) (with [Piet Van Mieghem](#))
- Liufei Yang, February – June 2021. [Developing a Markov-Modulated Process Model for Mobility Processes](#) (with [Piet Van Mieghem](#))

## VISITING PERIODS

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- Lyon, France, 4 – 8 June 2023. At Inria Lyon, working with [Laurent Pujo-Menjouet](#) and [Mostafa Adimy](#)
- Trento, Italy, 5 – 8 December 2022; 27 – 31 March 2023. At University of Trento, working with [Andrea Pugliese](#) and [Sara Sottile](#)
- Amsterdam and Groningen, the Netherlands, 21 – 25 November 2022. At Vrije Universiteit Amsterdam and Rijksuniversiteit Groningen, working with [Bob Rink](#) and [Hildeberto Jardón-Kojakhmetov](#)
- München, Germany, 15 April – 15 June 2019. At Technische Universität München (TUM), working with [Christian Kuehn](#) and [Hildeberto Jardón-Kojakhmetov](#)

## ORGANIZATION & SCIENTIFIC COMMITTEES

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Organizer</b> , “ <a href="#">Across generations in mathematical biology</a> ”, Trento.<br>Organizer of thematic workshop (planned).                                                                                                                                                                  | 8 – 9 October 2026      |
| <b>Minisymposium organizer</b> , <a href="#">ECMTB 2026</a> , Graz.<br>Minisymposium: MS79 “Heterogeneity in epidemic modelling”<br>Minisymposium: MS107 “Modeling Avian Influenza Dynamics”                                                                                                             | 13 – 17 July 2026       |
| <b>Seminar organizer</b> , University of Trento.<br>Jacobs Somnic, “ <a href="#">Stability Analysis of a Multistage Compartmental Model of Infection</a> ”; 28 January 2026.<br>Lukas Eigentler, “ <a href="#">Can we predict wavelength changes of banded vegetation patterns?</a> ”; 18 December 2025. | December 2025 – present |
| <b>Scientific committee</b> , <a href="#">Complex Networks 2024</a> , Istanbul, Turkey.<br>Member of the scientific committee which evaluates abstract and article submissions.                                                                                                                          | 10 – 12 December 2024   |
| <b>Minisymposium organizer</b> , <a href="#">ECMTB 2024</a> , Toledo.<br>Minisymposium: “Travelling wave phenomena in biology”                                                                                                                                                                           | 22 – 26 July 2024       |
| <b>Scientific committee</b> , <a href="#">Complex Networks 2023</a> , Menton Riviera, France.<br>Member of the scientific committee which evaluates abstract and article submissions.                                                                                                                    | 28 – 30 November 2023   |
| <b>Scientific committee</b> , <a href="#">FRCCS 2023</a> , Le Havre.<br>Member of the scientific committee which evaluates abstract and article submissions.                                                                                                                                             | 31 May – 02 June 2023   |
| <b>Minisymposium organizer</b> , <a href="#">ECMTB 2022</a> , Heidelberg.<br>Minisymposium: “Recent advances in mathematical modelling in neuroscience”                                                                                                                                                  | 19 – 23 September 2022  |
| <b>Seminar organizer</b> , Inria – Université Côte d'Azur.<br><a href="#">MathNeuro seminars</a> , cycle of seminars on mathematical models in neuroscience.                                                                                                                                             | April – September 2022  |
| <b>Organizer and scientific committee</b> , <a href="#">DSABNS 2020</a> , Trento.<br>Organizer and member of the scientific committee which evaluates abstract and article submissions.                                                                                                                  | 4 – 7 February 2020     |

## COMMUNICATIONS

- Invited speaker**, [A New Generation of Geometric Singular Perturbation Theory](#), Trento, Italy. 31 August – 4 September 2026  
Title: “*Geometric Singular Perturbation Theory in epidemic modelling: motivation and examples*”
- Contributed and invited speaker**, [ECMTB 2026](#), Graz. 13 – 17 July 2026  
Title (contributed): “*Avian Flu Modelling in Poultry Farms: Case Study of H5N1 HPAI in Northern Italy in 2021-2022*”.  
Title (invited): “*Slow-fast dynamics in a neurotransmitter release model: Delayed response to a time-dependent input signal*”.  
Part of MS101 “Advances in multiple timescale dynamics in neurons and related excitable systems”
- Invited speaker**, [Biomath 2026](#), Szeged, Hungary. 14 – 19 June 2026  
Title: “*Avian Flu Modelling in Poultry Farms: Case Study of H5N1 HPAI in Northern Italy in 2021-2022*”
- Contributed and invited speaker**, [DSABNS 2026](#), Granada, Spain. 2 – 6 February 2026  
Title (contributed): “*Avian Flu Modelling in Poultry Farms: Case Study of H5N1 HPAI in Northern Italy in 2021-2022*”.  
Title (invited): “*Slow-Fast Dynamics in a Neurotransmitter Release Model: Delayed Response to a Time-Dependent Input Signal*”.  
Part of Mini-symposium 02 “Nonlinear and Multiscale Brain Dynamics in Disease”
- Invited speaker**, [Biomath 2025](#), Sofia, Bulgaria. 15 – 20 June 2025  
Title: “*A general kinetic model for the spread of infectious diseases in continuously structured compartments*”
- Invited speaker**, [BIMSA Computational Math Seminar](#), Beijing. 22 May 2025  
Title: “*Geometric Singular Perturbation Theory in epidemic modelling: motivation and examples*”
- Invited speaker**, [CDLab](#), Udine. 1 April 2025  
Title: “*A general kinetic model for the spread of infectious diseases in continuously structured compartments*”
- Invited speaker**, [EPIMAT seminar](#), Trento. 11 March 2025  
Title: “*A general kinetic model for the spread of infectious diseases in continuously structured compartments*”
- Invited speaker**, [Numerical Aspects of Hyperbolic Balance Laws and Related Problems](#), Ferrara. 17 – 19 December 2024  
Title: “*A general kinetic model for the spread of infectious diseases in continuously structured compartments*”
- Invited speaker**, MACBES team, Inria d’Université Côte d’Azur. 18 November 2024  
Title: “*Various approaches to the mathematical modelling of epidemics*”
- Invited speaker**, [GIMC SIMAI YOUNG 2024](#), Napoli. 10 – 12 July 2024  
Title: “*A general kinetic model for the spread of infectious diseases in continuously structured compartments*”  
Part of the minisymposium “MS01 – Mathematical Models for Socio-Epidemiological Dynamics”
- Invited speaker**, Laboratoire de Mathématiques Appliquées du Havre. 2 May 2024  
Title: “*Various approaches to the mathematical modelling of epidemics*”
- Invited speaker**, [Integrated Mathematical approaches to Socio-Epidemiological Dynamics](#), Trento. 29 – 31 January 2024  
Title: “*A general kinetic model for the spread of infectious diseases in continuously structured compartments*”
- Poster presentation**, [Special Semester on Mathematical Methods in Medicine](#), Linz, Austria. 30 October – 3 November 2023  
Title: “*A general kinetic model for the spread of infectious diseases in continuously structured compartments*”.  
Part of workshop 1 “Epidemics modeling”
- Invited speaker**, [SIMAI 2023](#), Matera. 28 August – 1 September 2023  
Title: “*A general kinetic model for the spread of infectious diseases in continuously structured compartments*”.  
Part of the minisymposium “MS03: Recent Advances on the mathematical and numerical modeling of epidemics”
- Invited speaker**, Inria Lyon. 7 June 2023  
Title: “*Various approaches to the mathematical modelling of epidemics*”
- Contributed speaker**, [Workshop MSE](#), Napoli. 18 – 19 May 2023  
Title: “*A geometric analysis of the SIRS model with secondary infections*”
- Invited speaker**, University of Trento. 7 December 2022  
[Mathematics Seminar](#), title: “*Delayed loss of stability in multiple time scale models of natural phenomena*”
- Invited speaker**, Rijksuniversiteit Groningen. 23 November 2022  
[Floris Takens Seminar](#), title: “*Entry-exit functions in fast-slow systems with intersecting eigenvalues*”
- Invited speaker**, Vrije Universiteit Amsterdam. 21 November 2022  
[Extra Dynamics Seminar](#), title: “*A Geometric Singular Perturbation approach to epidemic compartmental models*”
- Invited speaker**, University of Edinburgh. 14 October 2022  
[Applied and Computational Mathematics](#), title: “*Delayed loss of stability in multiple time scale models of natural phenomena*”
- Contributed speaker**, [ECMTB 2022](#), Heidelberg. 19 – 23 September 2022  
Title: “*A generalization of the full SNARE-SM model*”.  
Part of the minisymposium “Recent advances in mathematical modelling in neuroscience”
- Contributed speaker**, [ENOC 2022](#), Lyon. 17 – 22 July 2022  
Title: “*Delayed loss of stability in multiple time scale models of natural phenomena*”.  
Part of the minisymposium “MS-05 Slow-Fast Systems and Phenomena”
- Contributed speaker**, [100 UMI - 800 UniPD](#), Padova. 23 – 27 May 2022  
Title: “*A Geometric Singular Perturbation approach to epidemic compartmental models*”
- Invited speaker**, University of Edinburgh. 11 March 2022  
[Edinburgh Dynamical Systems Study Group](#), title: “*Entry-exit functions: beyond eigenvalue separation*”
- Invited speaker**, University of Edinburgh. 18 June 2021  
[Edinburgh Dynamical Systems Study Group](#), title: “*A Geometric Singular Perturbation approach to epidemic compartmental models*”
- Contributed speaker**, [DSABNS 2020](#), Trento. 4 – 7 February 2020  
Title: “*A GSPT approach to epidemics on homogeneous graphs*”
- Invited speaker**, University of Trento. 12 September 2019  
[Doc in Progress](#), title: “*An introduction to Geometric Singular Perturbation Theory*”
- Contributed speaker**, [Edinburgh Slow-Fast-Ival](#), Edinburgh. 4 – 5 July 2019  
Title: “*A GSPT approach to perturbed SIR and SIRWS models*”

**Contributed speaker**, [DSABNS 2019](#), Naples.

3 – 6 February 2019

Title: “*A GSPT approach to perturbed SIR and SIRWS models*”

**Invited speaker**, Technische Universität München (TUM).

21 January 2019

[Oberseminar Dynamics](#), title: “*A GSPT approach to perturbed SIR and SIRWS models*”

## REVIEWING

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### Journals:

- [Acta Biotheoretica](#)
- [Advances in Difference Equations](#)
- [Applied Mathematical Modelling](#)
- [Bollettino dell'Unione Matematica Italiana](#)
- [Contemporary Mathematics](#)
- [Discrete and Continuous Dynamical Systems - Series B](#)
- [Electronic Research Archive](#)
- [Epidemiologia](#)
- [International Journal of Biomathematics](#)
- [Journal of Biological Dynamics](#)
- [Journal of Biological Systems](#)
- [Journal of Complex Networks](#)
- [Journal of Mathematical Biology](#)
- [Mathematical Biosciences and Engineering](#)
- [Mathematical Methods in the Applied Sciences](#)
- [Mathematical Reviews](#)
- [Mathematics](#)
- [Mathematics and Computers in Simulation](#)
- [Nonlinear Analysis: Hybrid Systems](#)
- [Nonlinear Analysis: Real World Applications](#)
- [Nonlinear Dynamics](#)
- [Physica D: Nonlinear Phenomena](#)
- [Proceedings of the Royal Society A](#)
- [Qualitative Theory of Dynamical Systems](#)
- [Results in Applied Mathematics](#)
- [SIAM Journal on Applied Dynamical Systems \(SIADS\)](#)

## MEMBERSHIP AND COLLABORATIONS

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### GNFM – INdAM

2025 – present

Member of the group [Gruppo Nazionale di Fisica Matematica](#), of the [Istituto Nazionale di Alta Matematica](#)

### EMS - TAG - MLS

2024 – present

Member of the [EMS Topical Activity Group Mathematical Modelling in Life Sciences](#) (EMS - TAG - MLS) of the [European Mathematical Society](#)

### ESMTB

2024 – present

Member of the [European Society for Mathematical and Theoretical Biology](#) (ESMTB)

### Collaborazioni Matematiche con il Sud Globale - UMI

2024 – present

Member of the group [Collaborazioni Matematiche con il Sud Globale](#) (Mathematical Collaborations with the Global South) of the [Unione Matematica Italiana](#)

### MSE - UMI

2023 – present

Member of the group [Modellistica Socio-Epidemiologica](#) (Social-Epidemiological Modelling) of the [Unione Matematica Italiana](#)

### CSSF

2023 – present

Member of the [Complex Systems Society France](#)

### Mathematical Epidemiology group, University of Trento

2021 – present

Collaborator of the [Mathematical Epidemiology group](#), University of Trento

### GNAMPA – INdAM

2017 – 2021

Member of the group [Gruppo Nazionale per l'Analisi Matematica, la Probabilità e le loro Applicazioni](#), of the [Istituto Nazionale di Alta Matematica](#)

## SOFTWARE

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MATLAB, MatCont, python, AUTO-07P, Wolfram Mathematica;  
L<sup>A</sup>T<sub>E</sub>X, Microsoft Office tools.

## LANGUAGES

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Italian (mother tongue), English (C1), French (basic).